Integrity Assurance of LIRTK using SS-RAIM against sensor faults for UAV applications

Noah Minchan Kim, Dongchan Min and Jiyun Lee
Korea Advanced Institute of Science and Technology (KAIST)

ABSTRACT

This study proposes an integrity assurance architecture of Loosely-Coupled Kalman Filter-based Inertial Aiding for RTK (LIRTK) using Solution Separation based Receiver Autonomous Integrity Monitor (SS-RAIM). An integrity risk allocation tree of LIRTK is developed for each sensor fault hypothesis including the nominal hypothesis, single-satellite fault hypotheses, an IMU sensor fault hypothesis, and an incorrect fix fault hypothesis. Two P(CF) requirements for integrity risk and false alarm rate were defined in order to calculate the Vertical Protection Level (VPL) of the fixed solution. In order to improve the fixed rate by lowering the P(CF) requirement for false alarm rate, an ambiguity check process is newly proposed. In the ambiguity check process, the fixed ambiguities estimated under nominal and fault hypothesis are compared. After passing the ambiguity check process, this study found a new lower bound that relaxes the P(CF) requirement for false alarm rate and validated its mathematical proof. VPL simulations were performed for different Global Navigation Satellite System (GNSS) measurement noise levels and Inertial Measurement Unit (IMU) sensor grades. The simulation results demonstrated that the ambiguity check process significantly improved the fixed rate of LIRTK and exhibited a much higher value compared to single-epoch RTK.

1. INTRODUCTION

Unmanned aerial vehicle (UAV) navigation systems require a higher level of accuracy and integrity as the level of automation increases and their operating range extends to more populated areas such as urban environments [1]. In this context, carrier-phase-based differential global navigation satellite system (CDGNSS) including Real-time kinematics (RTK) have been actively studied to obtain centimeter-level positioning accuracy for UAV applications [2, 3]. Integer ambiguity resolution is the key to achieving high-precision RTK solutions. In the process of fixing float ambiguities of carrier-phase measurements to integers, it is important to evaluate the correctness of the integer ambiguities because incorrectly fixed ambiguities would cause a bias in the position of users and consequently a threat to the navigation integrity. Therefore, RTK requires a high level of Probability of Correct Fix, or P(CF), to provide users with integer-fixed position solutions at the required level of integrity [4].

Methods to improve ambiguity resolution performance including raising P(CF) in the RTK have received a considerable amount of researcher attention. One of the most widely used methods is fusing RTK systems with other auxiliary sensors such as an inertial measurement unit (IMU). RTK integrated with Inertial navigation systems (INS) have been investigated in several studies for applications requiring rapid and accurate ambiguity resolution [5, 6]. In the INS-integrated RTK systems, IMU measurements are used to propagate the previous position to the current position, which constrains the search space of the ambiguities. The reduced ambiguity search space significantly improves ambiguity resolution performance, especially in GNSS-challenged areas [7, 8]. Furthermore, a tightly coupled Kalman Filter (TCKF)-based structure increases P(CF) significantly by using the integer ambiguities from the previous epoch to solve the integer ambiguities at the current epoch. However, when a cycle slip occurs, the true integer ambiguities change as well. Thus, the integer ambiguities in the state vector must be initialized correspondingly, and the integer ambiguities from the previous epoch are no longer in use. During UAV operations in urban areas, cycle slips will be highly likely to occur due to various causes such as high dynamic movement of GNSS antenna, momentary loss of phase lock, and multipath caused by the surrounding urban environments [9]. Thus, one must detect all cycle slips when using the TCKF-based RTK system. If cycle slips are missed to detect due to the measurement noise, etc., as a result, the integer ambiguities could be incorrectly fixed, which, in turn, causes a bias in the user's position and poses a threat to navigation integrity.

A single-epoch RTK can avoid the cycle slip problem since it estimates integer ambiguities at every single epoch using GNSS measurements received at a given single epoch. However, the single-epoch RTK shows smaller P(CF) than that of the TCKF-based RTK, indicating that the fixed rate is low. This result motivated us to develop the architecture of Loosely-Coupled Kalman Filter-based Inertial Aiding for RTK (LIRTK) for high-integrity CDGNSS [10]. In the LIRTK, single-epoch GNSS measurements are used to compute least-square float solutions not affected by cycle slips, and a time-propagated position solution obtained from the LCKF is used as a pseudo measurement in addition to the GNSS measurements to improve the P(CF). In [10], the integrity performance of LIRTK was evaluated in the vertical protection level (VPL) domain under fault-free conditions, and the results showed improved performance when compared to that of the single-epoch RTK.

The integrity of LIRTK must also be assured under sensor fault conditions by developing sensor fault monitors and quantifying the integrity risk against sensor faults. In order to evaluate the integrity risk of a CDGNSS that includes LIRTK, it is necessary to consider the P(CF), which is affected by sensor faults as well. When the exact size of the measurement fault is known, a previous study [11] was carried out to compute the P(CF) considering effects of the sensor faults. In [12], a preliminary study was performed to calculate P(CF) and cycle resolution integrity risk conservatively in the presence of bounded measurement errors and faults. These preliminary studies have a limitation in that they are applicable only under the assumption that the maximum size of sensor faults is known. In general, reference receiver faults lack established threat models and therefore cannot be bounded [13]. When the sensor fault is unbounded, it is possible to calculate the worst-case fault vector and worst-case P(CF) using the threshold of the fault monitor. However, when applying the worst-case fault during the computation of worst-case P(CF), the value of P(CF) may decrease sharply, possibly resulting in the failure to meet the integrity requirement and the inability to derive the protection level of the fixed solution.

In this study, we constructed an integrity assurance architecture for LIRTK based on Solution Separation based Receiver Autonomous Integrity Monitor (SS-RAIM) to mitigate the impact of faults on P(CF). By utilizing the distribution of fault-free navigation solutions and the threshold of fault monitors under fault hypothesis, SS-RAIM allows for the calculation of integrity risk conservatively without computing the P(CF) of an all-in-view solution affected by sensor faults. [13] performed a study on evaluating the integrity and continuity risk of single-epoch RTK in the presence of unbounded measurement faults by applying the SS-RAIM. In integrity assurance system of LIRTK compared to that of single-epoch RTK, requires additional considerations for IMU sensor faults and fault propagation through LCKF. Considering these factors, this study designed a integrity risk allocation tree of LIRTK and developed an SS-RAIM-based integrity assurance system. Furthermore, we propose an ambiguity check process that compares the ambiguity fixed under nominal hypothesis with the ambiguity fixed under fault hypothesis to improve the fixed rate.

This paper is organized as follows. Section 2 describes the basic system architecture of LIRTK. In Section 3, the integrity risk allocation tree of LIRTK and the SS-RAIM-based integrity assurance architecture for LIRTK are explained. In Section 4, a newly proposed ambiguity check process to alleviate the fixed rate is described. Finally, section 5 performs simulations of the proposed SS-RAIM-based integrity assurance architecture of LIRTK for various sensor noise levels and evaluates the performance compared to that of the single-epoch RTK.

2. SYSTEM ARCHITECTURE OF LIRTK

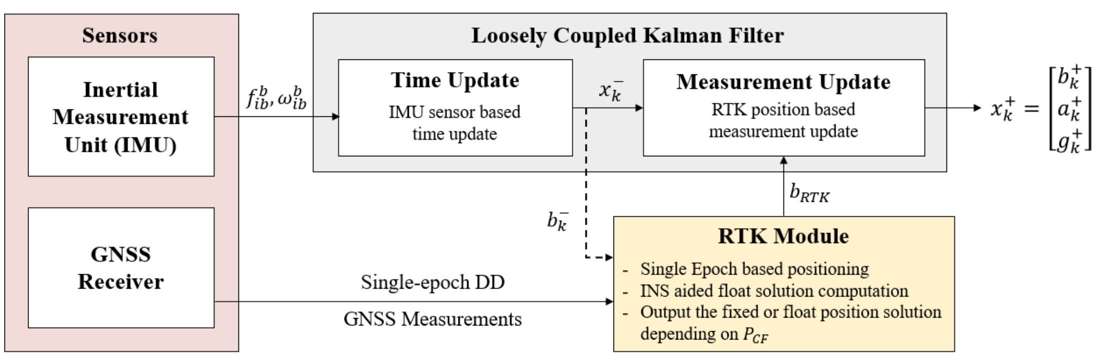

Fig. 1. Schematic diagram of LIRTK

Figure 1 illustrates the structure of LIRTK, an RTK/INS integrated navigation system based on LCKF [10]. The state of the LCKF is composed of position (i.e. a relative vector between the two receivers) which is represented as b, and inertial sensor biases which are represented as a for accelerometer and g for gyroscope in Fig. 1. The inertial measurement unit (IMU) sensor measurements $u_{IMU} = \left[f_{ib}^b, \omega_{ib}^b\right]^T$ are used as a control input in the time update step, and the position computed at the RTK module is used as a measurement input in the measurement update step to correct the INS-propagated state.

The RTK module of the LIRTK uses a least-squares filter based on the double-differenced (DD) GNSS measurements and the b_k^- which is the time-updated position from the INS to estimate the improved float ambiguity and float position solution compared to those of the single-epoch RTK without INS aiding. Once the float ambiguity solution is obtained, it is mapped into the integer number space to consider the integer constraints of the ambiguities. In LIRTK, the float ambiguity solution was mapped into the integer solution by the Integer Bootstrapping (IB) method because it enables us to compute the P(CF) in a straightforward manner [12].

After obtaining the integer solution of ambiguities, which is referred to as a fixed ambiguity solution, we have to decide on the acceptance of the fixed ambiguity solution. In [10], the acceptance of fixed ambiguity solution was determined by comparing P(CF) and the integrity risk allocated for the fault-free condition ($PHMI_{req,0}$), as only the fault-free condition was considered. In this study, an integrity assurance architecture for LIRTK based on SS-RAIM was designed. In detail, Section 3.1 introduces the integrity allocation tree of LIRTK, and Section 3.2 explains the process of deriving the fault-free solution for each fault hypothesis. In Sections 3.3 and 3.4, the process of calculating the false alarm rate and integrity risk, respectively, is explained. The derived P(CF) requirement conditions from each calculation process are also described. Finally, in Section 3.5, the proposed integrity assurance architecture is explained.

3. INTEGRITY ASSURANCE OF LIRTK USING SS-RAIM

3.1. Integrity risk allocation tree of LIRTK

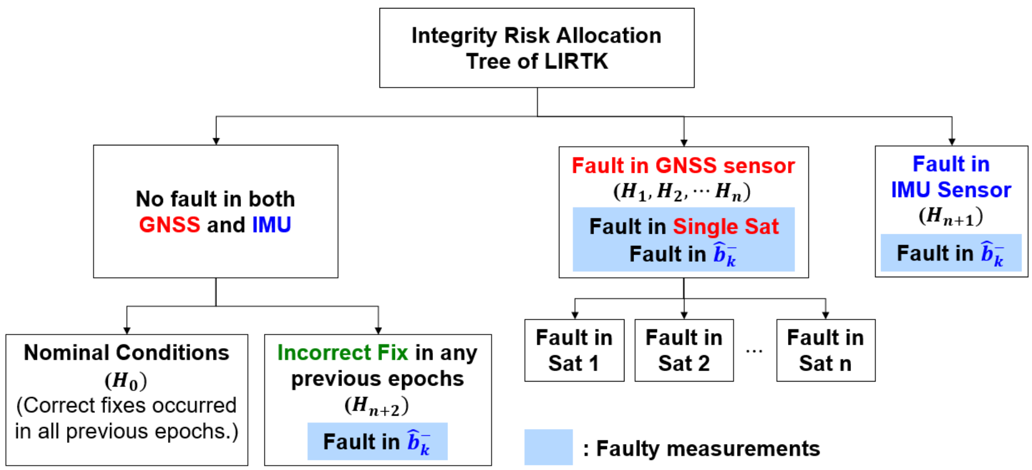

Fig. 1. Integrity risk allocation tree of LIRTK

In this study, we designed the integrity risk allocation tree of LIRTK, which is shown in the above figure. LIRTK generates the final output (fixed or float), by using the current GNSS measurements and $\hat{b}_k^-$ measurement that is time-propagated using the IMU measurements. This study only considers the single-satellite fault scenarios. Therefore, when receiving measurements from n satellites, there are n hypotheses of single-satellite fault and one hypothesis of IMU sensor fault, resulting in a total of n+1 hypotheses for the sensor measurement faults.

In CDGNSS, including LIRTK, there is a low probability of an Incorrect Fix occurring even in fault-free conditions, which can cause unbounded bias in the navigation solution. The VPL can bound the navigation error caused by an incorrect fix at the current epoch by considering the probability of incorrect fix, P(IF), during the integrity risk estimation process. Therefore, an incorrect fix that may occur at the current epoch is not a fault mode. If an incorrect fix occurred in the past, the bias included in the navigation solution would have propagated to the current estimate of $\hat{b}_k^-$ via LCKF. Hence, any instance of incorrect fix occurring in the past should be considered as a fault mode to ensure the integrity of LIRTK. Therefore, we defined a new fault hypothesis as the occurrence of an incorrect fix at least once in the past, even if no faults occurred in either the GNSS or IMU sensors. We refer to this fault hypothesis as the incorrect fix fault hypothesis. Accordingly, we defined the nominal condition as a state where no faults have occurred in any sensors and no incorrect fix has ever occurred in the past.

To summarize, if GNSS measurements are received from a total of n satellites, there are a total of n+2 fault hypotheses, including n single-satellite fault hypotheses $(H_1, H_2, \cdots H_n)$, one IMU sensor fault hypothesis (H_{n+1}), and one incorrect fix fault hypothesis (H_{n+2}). The probability of a single satellite fault was set to $10^{-5}/approach$, and under a fault condition, it was conservatively assumed that a fault occurred in the measurement of the single-satellite for all epochs from the first epoch to the current epoch.

Similarly, the probability of IMU sensor fault is also set per approach unit, and the same conservative assumption is applied that a fault occurs in every epoch. The equation below is used to update the prior probability of the incorrect fix fault hypothesis. In (1), $P(CF_0|H_0)_j$ represents the P(CF) of the all-in-view solution that is used as the measurement input for the LCKF at the j[th] epoch.

$$P_{H_{n+2}} = 1 - \prod_{j=1}^{k-1} P(CF_0|H_0)_j \tag{1}$$

3.2. Fault-free navigation solution for each fault hypothesis

To apply the integrity risk allocation tree defined in 3.1 to SS-RAIM, a fault-free navigation solution must be computed for each fault hypothesis. For each single-satellite fault hypothesis $(H_1, H_2, \cdots H_n)$, it is assumed conservatively that the fault was occurred at every epoch. Under this assumption, faults exist not only in the current single-satellite measurements but also faults from previous epochs are propagated to the $\hat{\mathrm{b}}_{\mathrm{k}}^{-}$ measurement via LCKF. Therefore, we obtained a fault-free navigation solution using only the GNSS measurements received at the current epoch except for the faulty satellite. (2) represents the linearized measurement equation for a fault-free navigation solution in an i[th] single-satellite fault hypothesis.

$$\begin{bmatrix} \nabla\Delta\rho_{k,i} \\ \nabla\Delta\phi_{k,i} \end{bmatrix} = H \begin{bmatrix} b \\ \nabla\Delta N \end{bmatrix} + \varepsilon \tag{2}$$

where $\nabla\Delta\rho_{k,i} \in \mathbb{R}^{n-2\times1}$ and $\nabla\Delta\phi_{k,i} \in \mathbb{R}^{n-2\times1}$ represent the received DD code and carrier measurements at the k[th] epoch from all satellites except the i[th] satellite. $b \in \mathbb{R}^{3\times1}$ and $\nabla\Delta N \in \mathbb{R}^{n-2\times1}$ are the true position and DD ambiguities, respectively. H is the observation matrix and ε is the measurement noise.

Assuming that faults occurred in all IMU sensor measurements from the first epoch to the current epoch, past IMU sensor faults propagate to the $\hat{\mathrm{b}}_{\mathrm{k}}^{-}$ measurement. Thus, to produce a fault-free navigation solution for the IMU sensor fault hypothesis (H_{n+1}), all GNSS measurements received at the current epoch were used except for the $\hat{\mathrm{b}}_{\mathrm{k}}^{-}$ measurement. The corresponding linearized measurement equation is shown below.

$$\begin{bmatrix} \nabla\Delta\rho_{k} \\ \nabla\Delta\phi_{k} \end{bmatrix} = H \begin{bmatrix} b \\ \nabla\Delta N \end{bmatrix} + \varepsilon \tag{3}$$

where $\nabla\Delta\rho_k \in \mathbb{R}^{n-1\times1}$ and $\nabla\Delta\phi_k \in \mathbb{R}^{n-1\times1}$ are the DD code and carrier phase measurements at the kth epoch, respectively.

In the incorrect fix fault hypothesis (H_{n+2}), the bias of navigation errors caused by past incorrect fixes is propagated to the current estimate, $\hat{\mathrm{b}}_{\mathrm{k}}^{-}$. Therefore, we computed the fault-free navigation solution by utilizing all current GNSS measurements except only for $\hat{\mathrm{b}}_{\mathrm{k}}^{-}$ measurement. Accordingly, the fault-free navigation solution of the incorrect fix fault hypothesis is identical to that of the IMU fault hypothesis, except for the different prior probabilities of the hypotheses.

We utilized the least-square method to solve the linearized measurement equation defined in each fault hypothesis, which resulted in the computation of the float position solution and float ambiguity solution. Subsequently, using the IB method described in section 2 for the all-in-view navigation solution, we calculated the fault-free navigation solution, $\check{b}_{i,fixed}$ and $P(CF_i|H_i)$.

In this paper, only the vertical component of the position error is utilized in all the derivations because it is typically the most demanding for precision aviation applications, and the lateral components can be handled similarly. The test statistic for each fault monitor of SS-RAIM was defined as the difference between the vertical component of the fault-free navigation solution obtained from each fault hypothesis and the vertical component of the all-in-view navigation solution as follows:

$$\Delta_i = \check{v}_{i,fixed} - \check{v}_{0,fixed} \tag{4}$$

The solution separation variance is given by following equation.

$$\sigma_{\Delta_i}^2 = \sigma_{\check{v}_{i,fixed}}^2 - \sigma_{\check{v}_{0,fixed}}^2 \tag{5}$$

3.3. False alarm rate evaluation

To detect faults using the test statistic derived in 3.2, an appropriate threshold must be set for each fault monitor. Typically, the threshold of each fault monitor in SS-RAIM is determined from the false alarm rate requirement and the distribution of the test statistic. The false alarm rate refers to the probability that a fault will be detected by the fault monitor in nominal condition. In SS-RAIM of a typical code-based navigation system, it is calculated as follows:

$$P_{FA,est} = \sum_{i=1}^{n_{sat}+2} P(|\Delta_i| > T_i|H_0) \cdot P_{H_0} \tag{6}$$

where T_i is the threshold of ith fault monitor in SS-RAIM. In the LIRTK, the equation above is expanded as follows, depending on whether the all-in-view navigation solution and fault-free navigation solution are correctly or incorrectly fixed.

$$P_{FA,est} = \sum_{i=1}^{n_{sat}+2} [P(|\Delta_i| > T_i|CF_0, CF_i, H_0) \cdot P(CF_0, CF_i|H_0) + P(|\Delta_i| > T_i|\{CF_0, CF_i\}^c, H_0) \cdot P(\{CF_0, CF_i\}^c|H_0)] \cdot P_{H_0} \tag{7}$$

When both the all-in-view navigation solution and the fault-free navigation solution are correctly fixed, the test statistic, Δ_i follows a zero-mean normal distribution, so it is easy to calculate the false alarm rate. Conversely, if either of the two navigation solutions is incorrectly fixed, it is difficult to calculate the false alarm rate accurately as the test statistic follows a biased normal distribution. Therefore, previous research took a conservative approach by assuming a false alarm rate of 1 in the case of an incorrect fix [13]. Thus, the overall false alarm rate equation was modified as follows:

$$P_{FA,est} = \sum_{i=1}^{n_{sat}+2} \left[P(|\Delta_i| > T_i|CF_0, CF_i, H_0) \cdot P(CF_0, CF_i|H_0) + \left(1 - P(CF_0, CF_i|H_0)\right)\right] \cdot P_{H_0} \tag{8}$$

The above equation is used to determine a threshold, T_i that sets $P_{FA,est}$ equal to $P_{FA,req}$. In order to find a positive threshold for all fault hypotheses, the sum of prior probabilities for incorrect fixes must be less than $P_{FA,req}$. This condition can be expressed as follows, and we refer to the sum of prior probabilities of incorrect fixes multiplied by P_{H0} included in (9) as "NC_Con", which is a necessary condition for Continuity risk.

$$\sum_{i=1}^{n_{sat}+2} \left(1 - P(CF_0, CF_i|H_0)\right) \cdot P_{H_0} < P_{FA,req} \tag{9}$$

To accurately calculate the $P(CF_0, CF_i|H_0)$, one must consider the strong correlation between CF_0 and CF_i. However, taking into account the correlation between the two events requires the consideration of a high-dimensional covariance matrix, leading to potential computational inefficiencies, and thus the lower bound of $P(CF_0, CF_i|H_0)$ was utilized in [13] to overbound the term of $1 - P(CF_0, CF_i|H_0)$ by assuming independence between the two events, as shown below:

$$P(CF_0, CF_i|H_0) \geq P(CF_0|H_0) \cdot P(CF_i|H_0) \tag{10}$$

$$\sum_{i=1}^{n_{sat}+2} \left(1 - P(CF_0|H_0) \cdot P(CF_i|H_0)\right) \cdot P_{H_0} < P_{FA,req} \tag{11}$$

If the inequality (11) is satisfied using the lower bound, it is possible to derive the threshold of each fault monitor for the fixed solution to be the final output of LIRTK. If the inequality (11) is not satisfied, a float solution is used as the final output. In this case, the threshold of each fault monitor for the float solution is derived to ensure that the value of $P_{FA,est}$ calculated using (12) is the same as $P_{FA,req}$.

$$P_{FA,est} = \sum_{i=1}^{n_{sat}+2} P\left(\left|\Delta_{i,float}\right| > T_{i,float} \middle| H_0\right) \cdot P_{H_0} \tag{12}$$

3.4. Integrity risk evaluation

Integrity risk in SS-RAIM refers to the probability that the error of the all-in-view navigation solution provided to the user exceeds the VPL under nominal and fault hypotheses, and is expressed as follows:

$$P_{HMI,est} = \sum_{i=0}^{n_{sat}+2} P(|\check{v}_0 - v| > \mathrm{VPL}|H_i) \cdot P_{H_i} \tag{13}$$

The all-in-view navigation solution under each fault hypothesis does not follow a zero-mean normal distribution because it includes faulty measurements. Consequently, the calculation of integrity risk using (13) is highly complicated. To address this complexity, [13] conservatively calculated integrity risk by using the fault-free navigation solutions instead of all-in-view navigation solutions under the fault hypotheses, as indicated by the inequality below.

$$|\check{v}_0 - v| = |\check{v}_0 - \check{v}_i + \check{v}_i - v| < |\check{v}_0 - \check{v}_i| + |\check{v}_i - v| < T_i + |\check{v}_i - v| \tag{14}$$

$$P_{HMI,est} \leq \sum_{i=0}^{n_{sat}+2} P(|\check{v}_i - v| + T_i > \mathrm{VPL}|H_i) \cdot P_{H_i} \tag{15}$$

When estimating integrity risk of LIRTK, it is important to consider whether the fault-free navigation solution is correctly fixed or incorrectly fixed. Similar to the calculation process for false alarm rate, we assume that integrity cannot be guaranteed when the fault-free navigation solution is incorrectly fixed. After dividing the integrity risk allocation equally between vertical and lateral components (with only vertical component shown in equation (15)) and considering that correct and incorrect fix events are mutually exclusive and exhaustive ($P(IF_i|H_i) = 1 - P(CF_i|H_i)$), the equation (15) can be expanded as follows:

$$P_{HMI,est} = \sum_{i=0}^{n_{sat}+2} \left[P(|\check{v}_i - v| + T_i > \mathrm{VPL}|CF_i, H_i) \cdot P(CF_i|H_i) + \frac{1}{2}P(IF_i|H_i)\right] \cdot P_{H_i} \tag{16}$$

The VPL is searched to achieve $P_{HMI,est}$ equal to $P_{HMI,req}$ by utilizing the threshold of each fault monitor and the distribution of fault-free navigation solutions. To calculate the appropriate VPL for the fixed solution using (16), the sum of the prior probabilities of incorrect fixes for each hypothesis must be less than $P_{HMI,req}$. This condition can be expressed as (17), and we refer to the sum of prior probabilities of incorrect fixes multiplied by P_{Hi} included in (17) as “NC_Int”, which is a necessary condition for Continuity risk.

$$\sum_{i=0}^{n_{sat}+2} \frac{1}{2}P(IF_i|H_i) \cdot P_{H_i} < P_{HMI,req} \tag{17}$$

If the equation (17) is satisfied, it is possible to calculate the VPL of the fixed solution. Otherwise, the final output of LIRTK is float solution, and VPL of float solution is calculated based on the following equation.

$$P_{HMI,est} \leq \sum_{i=0}^{n_{sat}+2} P\left(|\hat{v}_i - v| + T_{i,float} > \mathrm{VPL}\middle|H_i\right) \cdot P_{H_i} \tag{18}$$

3.5. Overall Integrity Assurance Architecture of LIRTK

The overall integrity assurance architecture of LIRTK is shown in Fig. 3. To ensure the integrity of fixed solutions, two P(CF) requirements for both integrity risk and false alarm rate must be satisfied. Otherwise, the final output will be a float solution, and the VPL of the float solution will be calculated via SS-RAIM.

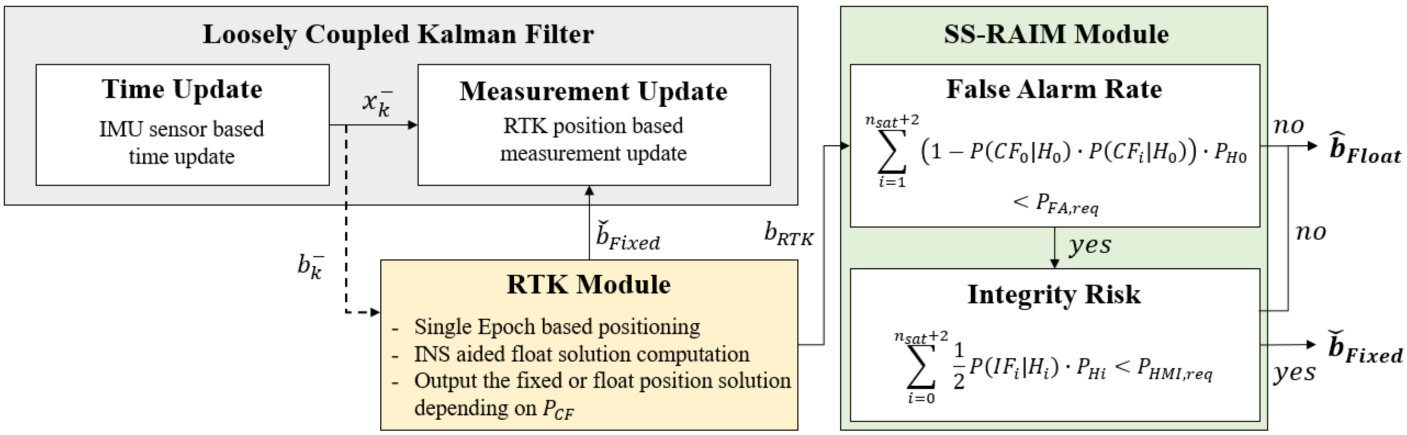

Fig. 3. Overall Integrity Assurance Architecture of LIRTK

4. AMBIGUITY CHECK PROCESS

4.1. P(CF) requirements for Integrity Risk and False Alarm Rate

To ensure the integrity of the fixed solution using SS-RAIM, both P(CF) requirements for integrity risk and false alarm rate must be satisfied. The value of $P(IF_i|H_i)$ in (17) is high because it is calculated using only the GNSS measurements of the current epoch. However, the overall value decreases as the prior probability of the fault hypothesis, P_{Hi}, is multiplied. On the other hand, applying the lower bound of (10) to the P(CF) requirement for false alarm rate yields the following inequality.

$$1 - P(CF_0|H_0) \cdot P(CF_i|H_0) > 1 - P(CF_i|H_0) = P(IF_i|H_0) = P(IF_i|H_i) \tag{19}$$

Therefore, when $P_{HMI,req}$ and $P_{FA,req}$ are comparable, satisfying the P(CF) requirement for false alarm rate is much more challenging due to the absence of multiplication of the prior probability of the fault hypothesis, while $1 - P(CF_0|H_0) \cdot P(CF_i|H_0)$ is greater than $P(IF_i|H_i)$. Based on the simulation results, loss of the fixed rate was observed due to the inability to meet the P(CF) requirements for false alarm rate with high GNSS noise levels. Therefore, this study proposes an ambiguity check process to improve the fixed rate by relaxing the P(CF) requirement for false alarm rate.

4.2. Concept of Ambiguity Check Process

In the ambiguity check process, the fixed ambiguities estimated under nominal and fault hypothesis are compared. We refer to the fixed ambiguities under nominal conditions as $\breve{a}_{0,fixed}$ and under fault hypothesis as $\breve{a}_{i,fixed}$. In the IMU fault (H_{n+1}) and incorrect fix fault (H_{n+2}) hypothesis, all GNSS measurements received at the current epoch are used to fix the ambiguities. If both $\breve{a}_{0,fixed}$ and $\breve{a}_{i,fixed}$ are correctly resolved, then the fixed ambiguity solutions for both hypotheses should be identical.

In the single-satellite fault hypotheses ($H_1, H_2, \cdots H_n$), the number of fixed ambiguities is smaller than nominal condition because the measurement of the faulty satellite is excluded when fixing the float ambiguities. Furthermore, if the reference satellite is faulty, it is impossible to make a direct comparison because the true solution of the fixed ambiguities will change. Instead, it is possible to convert the $\breve{a}_{0,fixed}$ into $\breve{a}_{i,fixed}$ domain through linear-transformation. We can easily set up a linear transformation matrix that converts from the $\breve{a}_{0,fixed}$ domain to the $\breve{a}_{i,fixed}$ domain since we know the information of the newly selected reference satellite after removing the faulty satellite. This linear transformation matrix is referred to as F_i. If the $\check{a}_{0,fixed}$ and the $\check{a}_{i,fixed}$ are correctly fixed, $F_i \cdot \check{a}_{0,fixed}$ and the $\check{a}_{i,fixed}$ should be the same.

Events where the linearly transformed $\check{a}_{0,fixed}$ is equivalent to $\check{a}_{i,fixed}$ from the ith fault hypothesis will be referred to as AC_i.

$$AC_i : F_i \cdot \check{a}_{0,fixed} = \check{a}_{i,fixed} \tag{20}$$

Given the condition AC_i, CF_0 is sufficient for CF_i and can be mathematically expressed as follows:

$$P(CF_i|CF_0, AC_i, H_0) = 1 \tag{21}$$

In this study, we utilized (21) to search for a lower bound with better performance than (10) under the given condition of AC_i, and we found a new lower bound as below equation. The detailed derivation for (22) is attached in Appendix A.

$$\frac{P(CF_0|H_0) \cdot P(CF_i|H_0)}{P(CF_0|H_0) \cdot P(CF_i|H_0) + P(IF_0|H_0)} < P(CF_0, CF_i|AC_i, H_0) \tag{22}$$

4.3. False alarm rate and Integrity risk evaluation after Ambiguity Check Process

If the ambiguity check process is not passed in the ith fault hypothesis, the probability of CF_0 and CF_i becomes zero. This indicates that an incorrect fix occurred during the estimation process of the fault-free navigation solution or the all-in-view navigation solution. Therefore, even if the ambiguity check process is passed in all remaining hypotheses except for the ith hypothesis, it is impossible to satisfy the P(CF) requirement for false alarm rate.

$$P(CF_0, CF_i|{AC_i}^c, H_0) = 0 \tag{23}$$

In this study, we set the fixed solution as the final output when the ambiguity check process was passed for all fault hypotheses. When the final output is a fixed solution after passing the ambiguity check process, the equations for false alarm rate are modified as (24). For integrity risk, it is possible to calculate integrity risk conservatively by following the same formula as (16). The detailed proofs for false alarm rate and integrity risk equation after ambiguity check process are attached in appendix B and C.

$$P_{FA,est} = \sum_{i=1}^{n_{sat}+2} \left[P(|\Delta_i| > T_i|CF_0, CF_i, H_0) \cdot P(CF_0, CF_i|H_0) + \left(1 - \frac{P(CF_0|H_0) \cdot P(CF_i|H_0)}{P(CF_0|H_0) \cdot P(CF_i|H_0) + P(IF_0|H_0)}\right)\right] \cdot P_{H_0} \tag{24}$$

As the false alarm rate equation is modified to (24), the "NC_Con" is also modified, as shown in (25).

$$\sum_{i=1}^{n_{sat}+2} \left(1 - \frac{P(CF_0|H_0) \cdot P(CF_i|H_0)}{P(CF_0|H_0) \cdot P(CF_i|H_0) + P(IF_0|H_0)}\right) \cdot P_{H_0} < P_{FA,req} \tag{25}$$

4.4. Overall Integrity Assurance Architecture of LIRTK with Ambiguity Check Process

The overall integrity assurance architecture of LIRTK with ambiguity check process is shown in Fig. 4. To calculate the VPL of the fixed solution in the end, both the P(CF) requirements for integrity risk and false alarm rate must be met, and the ambiguity check process must pass for all fault hypotheses. Otherwise, the final output will be a float solution, and the VPL of the float solution will be calculated via SS-RAIM. In addition to the architecture described in section 3.5, the VPL of the fixed solution can only be obtained by passing all ambiguity check processes. Therefore, there is a potential risk that the fixed rate may decrease if it fails to pass the ambiguity check process. However, the simulation results described in section 5 indicate that the failure to pass the ambiguity check process was infrequent, and instead, there was a significant decline in the fixed rate when the P(CF) requirement for false alarm rate in (11) was applied. Consequently, the utilization of the proposed ambiguity check process resulted in a significant improvement in the fixed rate. Further details are presented in Section 5.

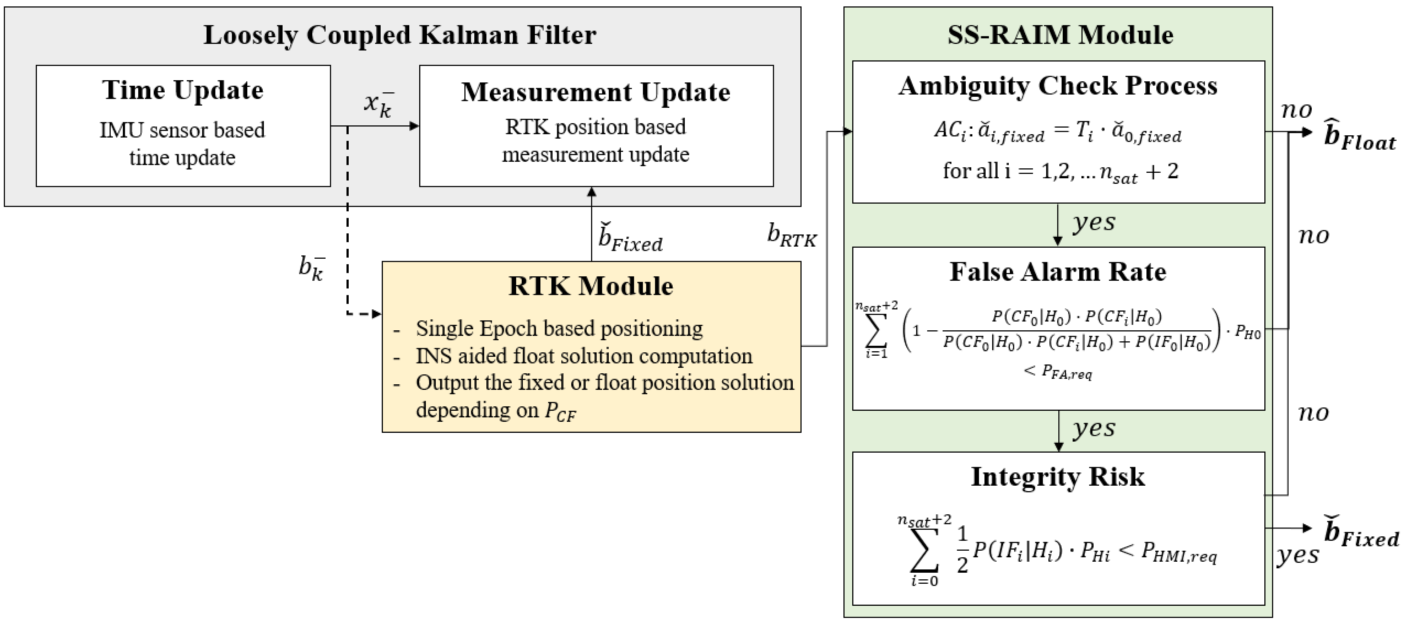

Fig. 4. Overall Integrity Assurance Architecture of LIRTK with Ambiguity Check Process

5. PERFORMANCE EVALUATION

To evaluate the performance of the proposed SS-RAIM based integrity assurance architecture of LIRTK with ambiguity check process, 24-hour VPL simulations were conducted for different sensor noise levels. The simulation parameters are listed in Table 1. σ_ρ and σ_ϕ represent the standard deviations of raw code and carrier phase measurement errors, respectively. The version of the AAD-B model scaled by using σ_ρ at a zenith direction as a reference was used for the code measurement error model. σ_ρ at an elevation angle of 90 degrees was approximately 17 cm for the AAD-B model. The AAD-B model was modified for different σ_ρ values of 25 ~ 35 cm at a zenith direction. For σ_ϕ, a constant error model was used, regardless of the elevation angles.

Table. 1. Simulation parameters for the VPL simulation

Parameter		Value
Requirements	$P_{HMI,req}$	10^{-7}
	$P_{FA,req}$	10^{-6}
IMU sensor	Grade	Consumer / Tactical
	Update rate	100 Hz
GNSS sensor	Constellation	24 GPS satellites [15] / 24 Galileo satellites [16]
	Frequency	GPS : L1 & L5 / Galileo : E1 & E5
	Multipath error model	LAAS AAD-B [17]
	σ_ρ (cm)	25 / 30 / 35
	σ_ϕ (mm)	5 / 6 / 7
	Update rate (Hz)	1

The uncertainty of the pseudo measurement, $\hat{b}_k^-$ used to estimate the float ambiguity and position solutions depends on the IMU sensor quality, which can be classified into aviation, tactical, and consumer grades. In this study, we compared the simulation results of tactical- and consumer-grade IMUs, and their noise levels are listed in Table 2.

Table2. Tactical-/Consumer- grade IMU parameters for the simulation [18]

IMU sensor							
Accelerometer				**Gyroscope**			
Bias PSD (m^2/s^5)		Noise PSD (m^2/s^3)		Bias PSD (rad^2/s^3)		Noise PSD (rad^2/s)	
Tactical	Consumer	Tactical	Consumer	Tactical	Consumer	Tactical	Consumer
10^{-7}	10^{-5}	0.002^2	0.2^2	$2{\cdot}10^{-12}$	$4{\cdot}10^{-11}$	0.00001^2	0.01^2

5.1. Performance of Ambiguity Check Process

To evaluate the performance of the proposed ambiguity check process, simulations were conducted for calculating NC_Con. When the ambiguity check process is not applied, the calculation of NC_Con is conducted using (11), whereas its application involves the utilization of (25). If the calculated NC_Con is higher than $P_{FA,req}$, the final output of LIRTK becomes a float solution, resulting in a loss of the fixed rate. Hence, we computed NC_Con using both equations and compared their performance in domain of the fixed rate. We conducted a 24-hour simulation considering the sensor noise level specified in Table 1. The performance analysis was conducted, disregarding the initial 10 epochs during which the covariance of LCKF did not converge.

Figure 5 presents the results of the NC_Con before and after the introduction of the ambiguity check process, when utilizing a consumer-grade IMU. In the absence of the ambiguity check process, the pass rate for the $P_{FA,req}$ threshold was 90.11% when σ_ϕ was 5mm, while it dropped to 0% for σ_ϕ values of 6mm and 7mm. After implementing the ambiguity check process, there was a significant improvement in the overall NC_Con. As a result, the pass rate for the $P_{FA,req}$ threshold at $\sigma_\phi = 5mm$ reached 100%, and there was a significant improvement in the pass rate to 84.24% when σ_ϕ was 6mm. In the case of σ_ϕ being 7mm, the overall NC_Con itself has significantly improved, but the pass rate was observed to be low at 0.19%. In addition, the increased noise level of GNSS measurements caused incorrect fixes during the computation of a fault-free navigation solution, leading to the failure of the ambiguity check process for a total of 40 epochs. Conversely, when σ_ϕ was 5mm and 6mm, no occurrences of failure in the ambiguity check process due to incorrect fixes were observed.

Figure 6 shows the results of the NC_Con when utilizing a tactical-grade IMU. When ambiguity check process was not introduced, the pass rates for $P_{FA,req}$ were found to be 90.16%, 0%, and 0% when σ_ϕ was 5mm, 6mm, and 7mm, respectively, which were identical to those of the consumer-grade IMU case. The reason why the identical results were observed regardless of the IMU grade is that if (11) is used to calculate the NC_Con, it depends on the $P(CF_i|H_0)$ derived from the fault hypothesis.. After introducing the Ambiguity check process, the new lower bound of $P(CF_0, CF_i|AC_i, H_0)$ proposed in this study showed significant variation in the results depending on the IMU grade. This is because if (25) is used to calculate the NC_Con, it relies on the $P(CF_0|H_0)$, which is determined from the all-in-view navigation solution estimation process. Therefore, after the application of the ambiguity check process, there was a significant improvement in the NC_Con compared to the consumer-grade IMU case, and the pass rate for the $P_{FA,req}$ threshold was observed to be 100% when σ_ϕ was 5mm and 6mm. When σ_ϕ was 7mm, similar to the consumer-grade IMU case, there were a total of 65 epochs where incorrect fix occurred, resulting in a loss of the fixed rate. However, since all remaining epochs satisfied the $P_{FA,req}$ threshold, the pass rate was very high at 99.93%, even when σ_ϕ was 7mm.

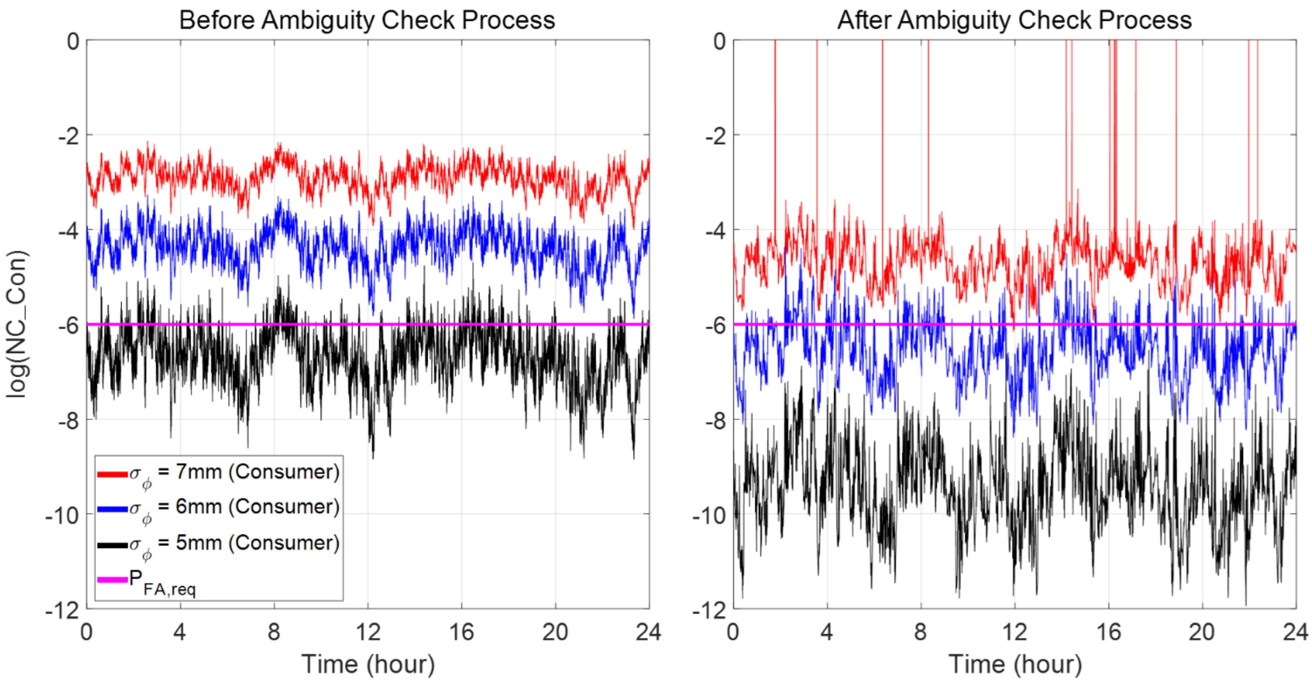

Fig. 5. 24-hour NC_Con simulation results of LIRTK with consumer-grade IMU

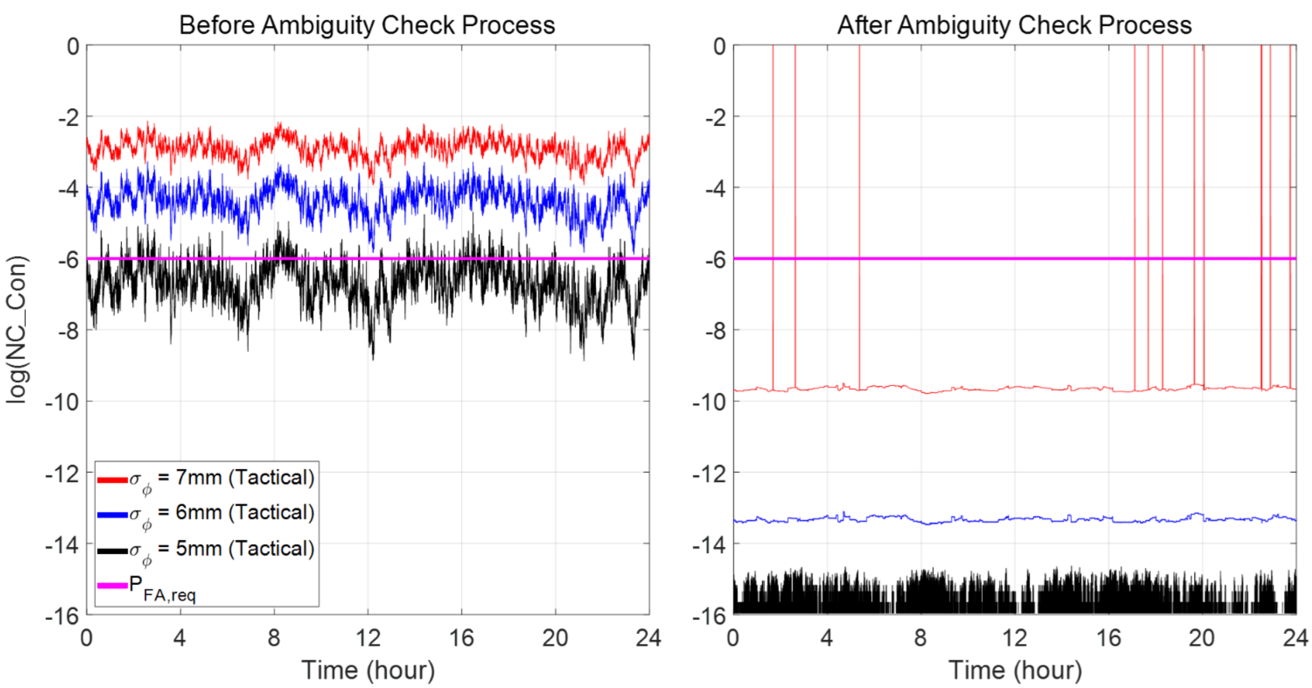

Fig. 6. 24-hour NC_Con simulation results of LIRTK with tactical-grade IMU

5.2. Vertical Protection Level Regarding the Sensor Noise Level

Fig. 7 and Fig. 8 show the three VPL simulation results; 1) the black line represents the single-epoch RTK; 2) the blue line represents the LIRTK with the consumer-grade IMU; and 3) the red line represents the LIRTK with the tactical-grade IMU for various GNSS noise levels. The VPLs for the float solution were at the meter level for all sensor noise levels, whereas the VPLs for the fixed solution were at the centimeter level. In the case of the single-epoch RTK, when σ_ϕ was 5mm, the fixed rate was observed to be 86.03%. However, as σ_ϕ increased, the P(CF) of the all-in-view solution decreased sharply, and at σ_ϕ=6mm and σ_ϕ=7mm, the fixed rate was 0%.

In the case of LIRTK with a consumer-grade IMU, the fixed rate was observed to be 100% at $\sigma_\phi = 5$mm. Additionally, when σ_ϕ was 6mm, there was a significant improvement in the fixed rate, reaching 84.24% compared to single-epoch RTK. However, when σ_ϕ was 7mm, it showed 0% fixed rate, similar to the single-epoch RTK.

When employing a tactical-grade IMU in LIRTK, the fixed rate was consistently 100% for both 5mm and 6mm values of σ_ϕ. Furthermore, even when σ_ϕ was 7mm, the fixed rate was remarkably high at 99.93%. As described in Section 5.1, the loss of fixed rate, which amounts to 0.07%, is caused by the occurrence of incorrect fixes during the fault-free navigation solution computation process, thereby failing to pass the ambiguity check process. The fault-free navigation solution is derived only using the current GNSS measurements, without utilizing the $\hat{b}_k^-$ measurement. Therefore, it is impossible to enhance the $P(CF_i|H_i)$ by utilizing the INS and as the noise level of GNSS measurements increases, $P(CF_i|H_i)$ decreases rapidly. Consequently, at σ_ϕ=7mm, $P(CF_i|H_i)$ was remarkably low, resulting in incorrect fixes occurring in a total of 65 epochs.

Overall, the fixed rate of the LIRTK was much higher compared to the single-epoch RTK. However, in the case of float solutions, the VPL of the LIRTK was higher compared to the single-epoch RTK. This is due to the conservative assumptions made regarding GNSS and IMU sensor faults, as described in the preceding Section 3.1 and 3.2. Under that conservative assumption, the fault-free navigation solution is derived only using the current GNSS measurements, without utilizing the $\hat{b}_k^-$ measurement. The VPL of SS-RAIM is ultimately determined by the threshold of the fault monitor and the distribution of the fault-free navigation solutions, as outlined in (18). While both the LIRTK and the single-epoch RTK share the same probability distribution of the fault-free navigation solutions, the inclusion of IMU fault (H_{n+1}) and incorrect fix fault (H_{n+2}) hypotheses in the LIRTK leads to a higher VPL for float solutions. On the other hand, the main strength of LIRTK lies in improving the P(CF) of the all-in-view solution. Consequently, from the perspective of VPL in the float solution, there may be a loss, but overall, the fixed rate will significantly increase..

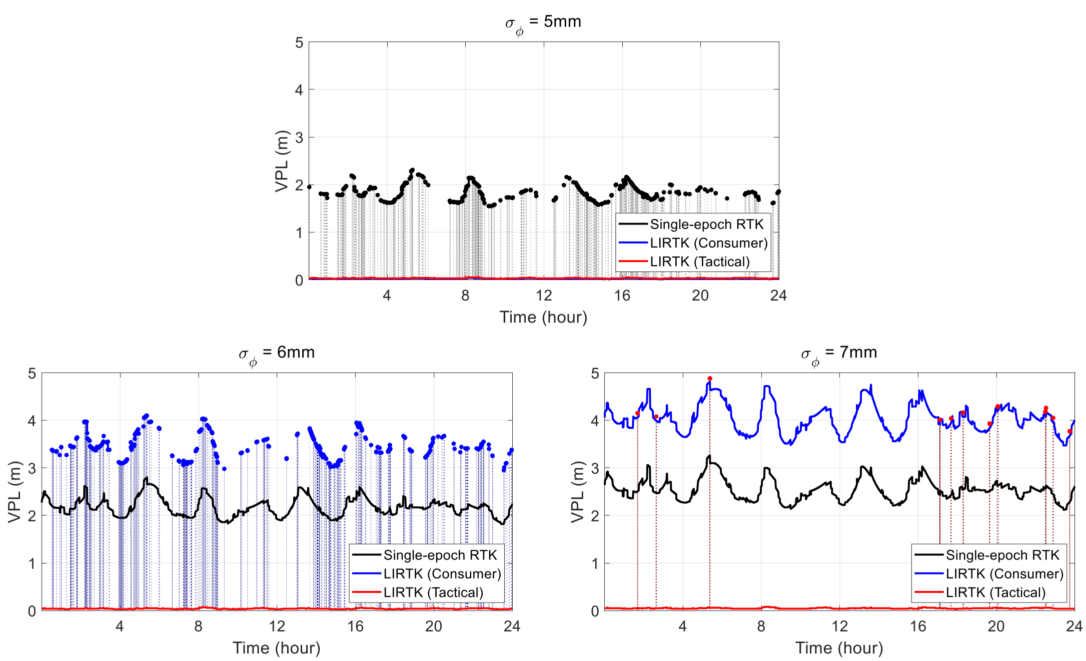

Fig. 7. 24-hour VPL simulation results of LIRTK and single-epoch RTK (Zoom-out)

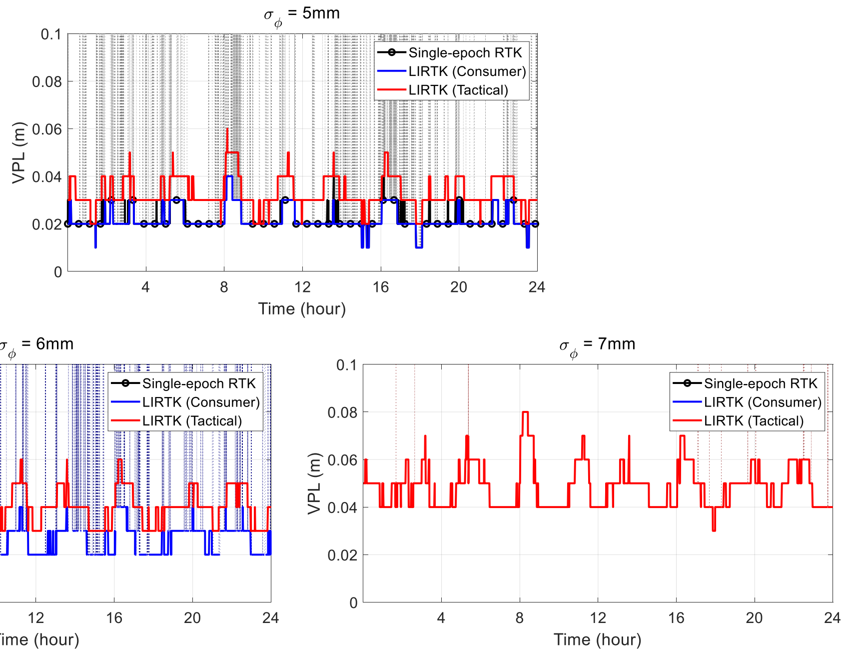

Fig. 8. 24-hour VPL simulation results of LIRTK and single-epoch RTK (Zoom-in)

CONCLUSIONS

This study proposes an integrity assurance architecture of LIRTK using SS-RAIM. An integrity risk allocation tree of LIRTK is developed for each sensor fault hypothesis including the nominal hypothesis, single-satellite fault hypotheses, an IMU sensor fault hypothesis and an incorrect fix fault hypothesis. Two P(CF) requirements for integrity risk and false alarm rate were defined in order to calculate the VPL of the fixed solution. In order to improve the fixed rate by relaxing the P(CF) requirement for false alarm rate, an ambiguity check process is newly proposed. After passing the ambiguity check process, this study found a new lower bound of $P(CF_0, CF_i|AC_i, H_0)$ that relaxes the P(CF) requirement for false alarm rate.

To evaluate the performance of the proposed ambiguity check process, simulations were conducted for calculating NC_Con. After introducing the Ambiguity check process, the new lower bound of $P(CF_0, CF_i|AC_i, H_0)$ proposed in this study showed significant variation in the results depending on the IMU grade. VPL simulations were performed for different GNSS measurement noise level and IMU sensor grades. According to the simulation results, the introduction of the ambiguity check process significantly improved the fixed rate of LIRTK. When employing a tactical-grade IMU in LIRTK, the fixed rate was consistently 100% for both 5mm and 6mm values of σ_ϕ. Furthermore, even when σ_ϕ was 7mm, the fixed rate was remarkably high at 99.93%.

Overall, the fixed rate of the LIRTK was much higher compared to the single-epoch RTK. However, in the case of float solutions, the VPL of the LIRTK was higher compared to the single-epoch RTK. To address these issues, one possible approach in the situation where LIRTK is the main system and single-epoch RTK is the sub-system, is to utilize the navigation solution of the sub-system when the final output of LIRTK is a float solution at that epoch.

ACKNOWLEDGMENT

Noah Minchan Kim was supported by Unmanned Vehicles Advanced Core Technology Research and Development Program through the National Research Foundation of Korea (NRF), Unmanned Vehicle Advanced Research Center (UVARC) funded by the Ministry of Science and ICT, the Republic of Korea (No. 2020M3C1C1A01086407).

REFERENCE

[1] International Civil Aviation Organization (ICAO). Unmanned Aircraft Systems (UAS); ICAO: Montreal, QC, Canada, 2011.

[2] Khalife, J. J., Bhattacharya, S., & Kassas, Z. M. (2018, September). Centimeter-accurate UAV navigation with cellular signals. In *Proceedings of the 31st International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2018)* (pp. 2321-2331).

[3] Ekaso, D., Nex, F., & Kerle, N. (2020). Accuracy assessment of real-time kinematics (RTK) measurements on unmanned aerial vehicles (UAV) for direct geo-referencing. *Geo-spatial information science*, *23*(2), 165-181.

[4] Pervan, B., & Chan, F. C. (2001, September). System concepts for cycle ambiguity resolution and verification for aircraft carrier landings. In *Proceedings of the 14th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GPS 2001)* (pp. 1228-1237).

[5] Skaloud, J. (1998, September). Reducing the GPS ambiguity search space by including inertial data. In *Proceedings of the 11th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GPS 1998)* (pp. 2073-2080).

[6] Han, H., Wang, J., Wang, J., & Tan, X. (2015). Performance analysis on carrier phase-based tightly-coupled GPS/BDS/INS integration in GNSS degraded and denied environments. *Sensors*, *15*(4), 8685-8711.

[7] Scherzinger, B. M. (2002, January). Robust positioning with single frequency inertially aided RTK. In *Proceedings of the 2002 National Technical Meeting of the Institute of Navigation* (pp. 911-917).

[8] Lee, H. K., Wang, J. I. N. L. I. N. G., & Rizos, C. H. R. I. S. (2005). An integer ambiguity resolution procedure for GPS/pseudolite/INS integration. *Journal of Geodesy*, *79*, 242-255.

[9] Shamaei, K., & Kassas, Z. M. (2019, September). Sub-meter accurate UAV navigation and cycle slip detection with LTE carrier phase measurements. In *Proceedings of the 32nd international technical meeting of the satellite division of the institute of navigation (ION GNSS+ 2019)* (pp. 2469-2479).

[10] Kim, N. M., Min, D., & Lee, J. (2022, January). Correct Fix Probability Improvement Method via INS Aiding to Single Epoch RTK System. In *Proceedings of the 2022 International Technical Meeting of The Institute of Navigation* (pp. 1304-1319).

[11] Teunissen, P. J. G. (2001). Integer estimation in the presence of biases. *Journal of Geodesy*, *75*, 399-407.

[12] Khanafseh, S., Joerger, M., & Pervan, B. (2012, April). Integrity risk of cycle resolution in the presence of bounded faults. In *Proceedings of the 2012 IEEE/ION Position, Location and Navigation Symposium* (pp. 664-672). IEEE.

[13] Khanafseh, S., Langel, S., Joerger, M., & Pervan, B. (2013, September). Evaluating Integrity and Continuity Risks of Cycle Resolution in the Presence of Receiver Faults. In *Proceedings of the 26th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2013)* (pp. 2583-2591).

[14] Teunissen, P. J. (2001, June). GNSS ambiguity bootstrapping: theory and application. In *Proceedings of international symposium on kinematic systems in Geodesy, geomatics and navigation* (pp. 246-254).

[15] SC, R. (2001). Minimum Operational Performance Standards for Global Positioning System/Wide Area Augmentation System Airborne Equipment. *RTCA/DO-229C*.

[16] Wallner, S., & Boyero, J. P. (2014). Personal conversation within the ARAIM working group C.

[17] RTCA, S. (2004). Minimum aviation system performance standard for the local area augmentation system (LAAS). *RTCA DO-245A*.

[18] Groves, P. D. (2015). Principles of GNSS, inertial, and multisensor integrated navigation systems, [Book review]. *IEEE Aerospace and Electronic Systems Magazine*, *30*(2), 26-27.

APPENDIX A. DERIVATION OF LOWER BOUND OF $\boldsymbol{P(CF_0, CF_i|AC_i, H_0)}$

By using the definition of the conditional probability, $P(CF_0, CF_i|AC_i, H_0)$ is expanded as follows:

$$P(CF_0, CF_i|AC_i, H_0) = P(CF_i|CF_0, AC_i, H_0) \cdot P(CF_0|AC_i, H_0). \tag{A.1}$$

If AC_i is given condition, CF_0 becomes a sufficient condition for CF_i.

$$P(CF_i|CF_0, AC_i, H_0) = 1 \rightarrow P(CF_0, CF_i|AC_i, H_0) = P(CF_0|AC_i, H_0) = \frac{P(CF_0, AC_i|H_0)}{P(AC_i|H_0)} \tag{A.2}$$

Because CF_0 and IF_0 are mutually exclusive and collectively exclusive events, $P(AC_i|H_0)$ is expanded as follows:

$$P(AC_i|H_0) = P(CF_0, AC_i|H_0) + P(IF_0, AC_i|H_0). \tag{A.3}$$

Based on the facts that $P(AC_i|CF_0, CF_i, H_0) = 1$ and $P(CF_i|CF_0, AC_i, H_0) = 1$, the following holds true:

$$P(CF_0, AC_i|H_0) = P(CF_0, CF_i|H_0). \tag{A.4}$$

Substituting (A.4) and (A.3) into (A.2), we obtain

$$P(CF_0, CF_i|AC_i, H_0) = \frac{P(CF_0, CF_i|H_0)}{P(CF_0, CF_i|H_0) + P(IF_0, AC_i|H_0)} \tag{A.5}$$

so that

$$P(CF_0, CF_i|AC_i, H_0) > \frac{P(CF_0, CF_i|H_0)}{P(CF_0, CF_i|H_0) + P(IF_0|H_0)} = \frac{1}{1 + \frac{P(IF_0|H_0)}{P(CF_0, CF_i|H_0)}}. \tag{A.6}$$

In (A.6), the lower bound of $P(CF_0, CF_i|AC_i, H_0)$ is minimized when $P(CF_0, CF_i|H_0)$ is minimized. We already defined the lower bound of $P(CF_0, CF_i|H_0)$ in (10). Finally, substituting (10) into (A.6), new lower bound of $P(CF_0, CF_i|AC_i, H_0)$ is derived.

$$P(CF_0, CF_i|AC_i, H_0) > \frac{P(CF_0|H_0) \cdot P(CF_i|H_0)}{P(CF_0|H_0) \cdot P(CF_i|H_0) + P(IF_0|H_0)} \tag{A.7}$$

APPENDIX B. FALSE ALARM RATE EVALUATION AFTER AMBIGUITY CHECK PROCESS

The false alarm rate of the fixed solution is calculated only when the ambiguity check process has been passed for all fault hypotheses. Accordingly, the formula for calculating the false alarm rate is modified as follows:

$$P_{FA,est} = \sum_{i=1}^{n_{sat}+2} P(|\Delta_i| > T_i, AC_i|H_0) \cdot P_{H_0}. \tag{B.1}$$

Similar to (7), (B.1) is expanded as follows:

$$P_{FA,est} = \sum_{i=1}^{n_{sat}+2} P(|\Delta_i| > T_i, AC_i, CF_0, CF_i|H_0) \cdot P_{H_0} + \sum_{i=1}^{n_{sat}+2} P(|\Delta_i| > T_i, AC_i, \{CF_0, CF_i\}^c|H_0) \cdot P_{H_0}. \tag{B.2}$$

By using the definition of conditional probability, $P(|\Delta_i| > T_i, AC_i, CF_0, CF_i|H_0)$ can be conservatively estimated as

$$\begin{aligned} &P(|\Delta_i| > T_i, AC_i, CF_0, CF_i|H_0) \\ &\quad = P(|\Delta_i| > T_i, AC_i|CF_0, CF_i, H_0) \cdot P(CF_0, CF_i|H_0) \\ &\quad < P(|\Delta_i| > T_i|CF_0, CF_i, H_0) \cdot P(CF_0, CF_i|H_0). \end{aligned} \tag{B.3}$$

By applying the same logic in (B.3), $P(|\Delta_i| > T_i, AC_i, \{CF_0, CF_i\}^c|H_0)$ can be conservatively estimated as

$$\begin{aligned} &P(|\Delta_i| > T_i, AC_i, \{CF_0, CF_i\}^c|H_0) \\ &\quad = P(|\Delta_i| > T_i|AC_i, \{CF_0, CF_i\}^c, H_0) \cdot P(\{CF_0, CF_i\}^c|AC_i, H_0) \cdot P(AC_i|H_0) \\ &\quad < P(\{CF_0, CF_i\}^c|AC_i, H_0) \\ &\quad < 1 - \underline{P(CF_0, CF_i|AC_i, H_0)}, \end{aligned} \tag{B.4}$$

where $\underline{P(CF_0, CF_i|AC_i, H_0)}$ is the lower bound of $P(CF_0, CF_i|AC_i, H_0)$, and we already derive that lower bound in Appendix A. Substituting (A.7), (B.3), (B.4) into (B.2), the false alarm rate evaluation equation after ambiguity check process is derived as follows.

$$P_{FA,est} = \sum_{i=1}^{n_{sat}+2} \left[P(|\Delta_i| > T_i|CF_0, CF_i, H_0) \cdot P(CF_0, CF_i|H_0) + \left(1 - \frac{P(CF_0|H_0) \cdot P(CF_i|H_0)}{P(CF_0|H_0) \cdot P(CF_i|H_0) + P(IF_0|H_0)}\right)\right] \cdot P_{H_0} \tag{B.5}$$

APPENDIX C. INTEGRITY RISK EVALUATION AFTER AMBIGUTIY CHECK PROCESS

Integrity risk evaluation formula after ambiguity check process is given by:

$$P_{HMI,est} = P(|v_0 - v| > VPL|H_0) \cdot P_{H_0} + \sum_{i=1}^{n_{sat}+2} P(|v_0 - v| > VPL, |\Delta_i| > T_i, AC_i|H_i) \cdot P_{H_i}. \tag{C.1}$$

For convenience, $P(|v_0 - v| > VPL, |\Delta_i| > T_i, AC_i|H_0)$ can be conservatively estimated as

$$\begin{aligned} &P(|v_0 - v| > VPL, |\Delta_i| > T_i, AC_i|H_0) \\ &\quad < P(|v_0 - v| > VPL, |\Delta_i| > T_i|H_0) \\ &\quad < P(|\check{v}_i - v| + T_i > VPL|H_i). \end{aligned} \tag{C.2}$$

Let us define $T_0 = 0$ and substitute (C.2) into (C.1), integrity risk evaluation formula becomes

$$P_{HMI,est} = \sum_{i=0}^{n_{sat}+2} P(|v_i - v| + T_i > VPL|H_i) \cdot P_{H_i} \tag{C.3}$$

and this is identical to the (15) used before the ambiguity check process.